

Jacob Furtaw

jfurtaw97@gmail.com | 410-533-7663 | www.linkedin.com/in/jacob-furtaw/ | www.jfcoded.com/projects
Baltimore, MD | Willing and Ready to Relocate Anywhere

Professional Summary

Results-driven ML Engineer & Agentic AI Builder with deep expertise in autonomous agents, multi-agent orchestration, and tool-calling systems. I specialize in transforming ambiguous requirements into production-ready agentic solutions through custom tool development, detailed system prompts, persistent memory/context management, and rigorous iteration — delivering reliable, observable systems that actually work in complex environments.

Skills

Agentic AI & Systems: Autonomous Agents • Multi-Agent Orchestration • Tool Calling & Reasoning Loops • Memory Management & Architectures • Advanced Prompt Engineering • RAG Pipelines • Observability & Transparency

Models & Inference: OpenAI, Anthropic, Grok, Gemini APIs • Local/Open-Weight Models (Ollama, vLLM, Llama.cpp, Nemotron Nano 3 Omni, Qwen2.5-Omni, Gemma) • Multimodal (Vision/Audio) • Quantization • Fine-Tuning

Frameworks & Tools: LangChain, Llama-Index, FastAPI, PyTorch, Transformers, Hugging Face, Docker, Git, Jupyter, NVIDIA NIMs & Foundation Models

Cloud & Deployment: AWS (MLOps Specialization in Progress) • Local NVIDIA Inference (DGX/Blackwell) • Multi-Cloud Ready

Work Experience

GenAI Engineer | Stealth AI Startup | Remote January 2026 – Present

- Designing and building **production-grade local multi-agent systems** with advanced tool calling, memory management, and RAG for Family and Multi-Family Office clients, delivering reliable, privacy-first automation for complex financial and operational workflows.

Co-Founder & Machine Learning Engineer | SurgePoint Software | Remote August 2023 – January 2025

- Led development of a production **RAG-powered AI student assistant** (Llama 3.1 / Mistral) serving 15k+ students with advanced prompt engineering and reliable agent performance.
- Built a **custom RAG pipeline** (Milvus / ChromaDB) and curated a high-quality **13M-line dataset** from 200M+ raw documents — improved semantic relevance by **50-75%** and significantly reduced hallucinations.
- Collaborated with a 6-person cross-functional team in an agile environment; delivered an MVP that **won \$1,000** in the Towson Startup Pitch Competition.

Advanced Repair Agent | Geek Squad | On-Site March 2022 – April 2026

- Consistently **top-performing** Advanced Repair Agent across the marketplace (6 stores); averaged **1.2+ tags/hour and 150%+ utilization** while resolving hundreds of complex issues monthly.
- Designed and implemented operational improvements that increased overall team productivity by **>50%**, earning formal recognition from management.

Research Projects

Cloak AI - Privacy First Local AI Agent

- Built a fully local multi-model AI agent with genuine tool use, mature RAG, dynamic model switching, and multimodal (text/image/audio) capabilities. Runs on consumer hardware.
- Implemented **inline real-time agent transparency/observability** for reasoning & tool-call monitoring
- Tech Stack:** React 19 + TypeScript, Python/FastAPI, Ollama, Nemotron Nano 3 Omni, NVIDIA Nemotron Parse. Features include user-created knowledge bases, an advanced web search tool, and inline image & stock data display.

Chat RAG & Supporting Projects | [Project Link](#)

- Chat RAG:** Advanced modular RAG coding assistant supporting hundreds of local (Ollama/Hugging Face) and API-based models (OpenAI, Anthropic, NVIDIA). Includes dynamic model switching, custom prompts, quantization, and flexible data ingestion from local files, GitHub, and vector databases (**25 stars on GitHub**).
- Agent Qwen:** Multimodal agent (Qwen2.5-Omni-7B) with voice I/O + image/video — Gradio, Whisper, PyTorch.

Automatic Identification of Equivalent Mutants using an ASTNN(GNN) | [Project Link](#)

- Delivered a transformer-based binary classification model (CodeBERT) as part of a 5-person Scrum team, improving F1 score and accuracy from 79% to 92% through hyperparameter tuning and dataset balancing techniques.

- Applied grid search and random search for hyperparameter optimization, along with over/undersampling methods to address class imbalance.

Education

Bachelor of Science in Computer Science, Software Engineering Concentration

December 2023

Towson University, Towson, MD

Clubs: Machine Learning Research Group

Coursework: Data Structures & Algorithms, Software Engineering, Databases, Linear Algebra, Statistics, Artificial Intelligence,